

~~$\therefore$ By Theorem 2, the system of equations corresponding to $Ax=b$ is consistent, i.e. there exists a solution set (Wrong)~~

$\therefore$ By Theorem 2, the augmented matrix $[a_1 \ a_2 \ \dots \ a_n \ b]$ corresponds to a consistent system of equations, i.e. there exists a solution set.

By Theorem 3, the system of equations $Ax=b$ has the same solution set as $[a_1 \ a_2 \ \dots \ a_n \ b]$.

$\therefore Ax=b$ has a solution. Notice, we have not chosen any particular b .

$\therefore Ax=b$ has a solution for each b in $\mathbb{R}^m$.

Suppose, (a) be True. $\therefore$ For each b in $\mathbb{R}^m$, the equation $Ax=b$ has a soln. Let A be an $m \times n$ matrix with columns $[a_1 \ a_2 \ \dots \ a_n]$. By Theorem 3, $Ax=b$ has the same solution as the vector equation $x_1 a_1 + x_2 a_2 + \dots + x_n a_n = b$ which in turn has the same solution set as the system of equations whose corresponding augmented matrix is $[a_1 \ a_2 \ \dots \ a_n \ b]$.

$\therefore$ The augmented matrix $[a_1 \ a_2 \ \dots \ a_n \ b]$ has a solution set. We can reduce this matrix to ~~an~~ ^{reduced} echelon form by the Row Reduction Algorithm. Let that matrix be E . ~~From Theorem 2, there is no row in the matrix of the form $[0 \ 0 \ \dots \ 0 \ b]$ where $b \neq 0$.~~

~~$\therefore$ In any echelon form of A , there are no zero rows. So~~

By Theorem 1, matrix E is one and unique. From Theorem 2, there is no row in the matrix of the form $[0 \ 0 \ \dots \ 0 \ b]$ where $b \neq 0$.

$\therefore$ In the reduced echelon form of A , there are no zero rows. So, there are at least one non-zero element in each row. By the defn. of reduced echelon form, the leading entry of each nonzero row is 1. By defn, these leading 1's are pivot positions. $\therefore$ Each row of the matrix A has a pivot position. So, (a) and (d) are equivalent $\square$